

AL AKHAWAYN UNIVERSITY in IFRANE

School of Science & Engineering

CSC5356 - Data Engineering and Visualization

Final project

Spring 2026

In this final project, you will have to build on the enterprise-grade pipeline of assignment #1, by:

1. Changing the storage layer to MinIO an S3 compatible blob storage (3 nodes of possible)
2. Adopt Parquet as a storage format and control
3. Deploy Apache Iceberg as a **Transactional Metadata Layer with a catalog**
4. Choose a processing engine and deploy it (Spark, Trino, and Flink)
5. **Query tool:** Trino or Athena
6. Write and execute analytics queries in SQL over different periods of time

Careful attention need to be given to 1. partitioning and to 2. the appropriate configuration of the sink properties, especially flush.size and time-based rotation strategies in order to ensure balancing file size, processing latency, and delivery guarantees (exactly-once vs. at-least-once)